

# V-Model

Model that relates specification and development activities to corresponding test levels.

## Core idea

Each definition/development stage has a related test activity, reinforcing early test planning.

## Official site and documentation

Use the official V-Model source as the primary reference: <https://www.istqb.org/>. Official documentation should be the first source for installation, supported versions, syntax changes and new features.

## Test levels

Component, Integration, System and Acceptance Testing can map to different specification levels.

## Traceability

Requirements and tests can be linked to demonstrate coverage and change impact.

## QA use

The model reinforces that testing should be planned while requirements and architecture are defined.

## When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

## How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

## What it is and where it fits

V-Model is a tool or technology that can support Quality Engineering work. Before adopting it, understand the problem it solves, where it fits in the workflow and what evidence or productivity gain it should provide.

## First QA-oriented use

Pick one small, verifiable task in V-Model, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

## Using it in a real project

Define where V-Model fits, who owns configuration, how credentials and data are handled and how results are shared. Integrate it into the team workflow instead of creating an isolated activity.

## **Adoption good practices**

Standardize versions and configuration, document important decisions, keep secrets out of code and preserve reproducible examples. Reassess whether V-Model still delivers value as the product evolves.

## **Common issues and troubleshooting**

When V-Model fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.